COMPUTER NETWORKS

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BACHELORS OF TECHNOLOGY

CSE

To

K.R Mangalam University

by

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Introduction

This experiment uses Cisco Packet Tracer's Simulation Mode to visually trace how packets travel between devices in a small Ethernet LAN. A simple LAN was constructed using one Cisco 2960 switch and three PCs (PC0, PC1, PC2) connected via Copper Straight-Through cables, all sharing the 192.168.20.0/24 subnet. The primary goals of this experiment are to observe the ARP broadcast and unicast reply sequence before an ICMP ping is sent, to compare packet counts between the first ping (which involves ARP resolution) and the second ping (which uses the ARP cache), to understand how a switch dynamically builds its MAC address table from live traffic, and to interpret the Simulation Mode Event List timeline step by step.

Before analyzing each task individually, it is essential to understand the interaction between ARP and ICMP in an Ethernet LAN. When a device attempts to send an ICMP ping to a destination for the first time, it has no knowledge of the destination's MAC address. ARP (Address Resolution Protocol) is triggered to resolve the IP address to a MAC address through a broadcast frame. The switch, upon receiving this broadcast, floods the frame to all connected ports and simultaneously learns the source device's MAC address. Once ARP is resolved, ICMP Echo Requests are sent as unicast frames directly to the destination port. On subsequent pings, the ARP cache prevents the broadcast from being re-sent, resulting in faster and more efficient communication.

Network Topology

A simple LAN consisting of one Cisco 2960 switch and three PCs connected via Copper Straight-Through cables. All devices share the 192.168.20.0/24 subnet.

Device	Interface	IP Address	Subnet Mask	Connected To
PC0	FastEthernet0	192.168.20.1	255.255.255.0	Switch0 Fa0/1
PC1	FastEthernet0	192.168.20.2	255.255.255.0	Switch0 Fa0/2
PC2	FastEthernet0	192.168.20.3	255.255.255.0	Switch0 Fa0/3
Switch0	Fa0/1–Fa0/3	- (Layer 2)	-	All three PCs

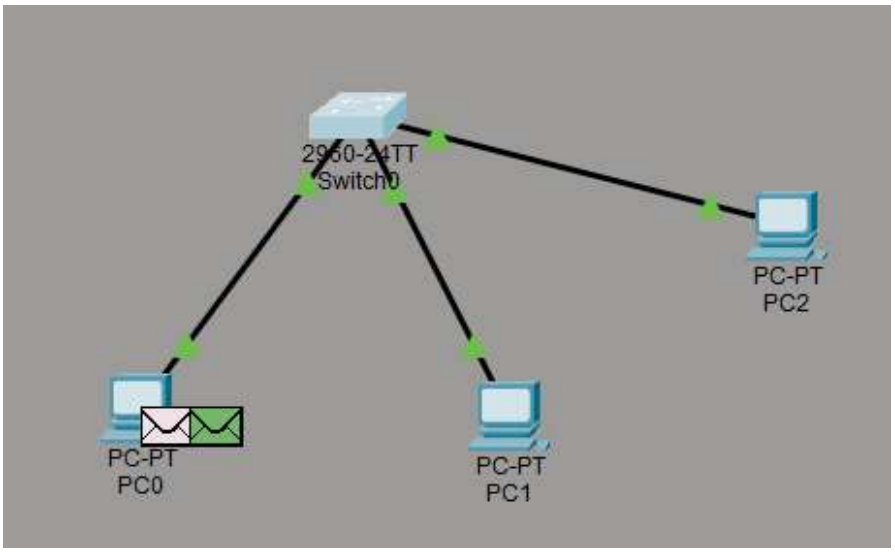

Fig 1: Network topology in Packet Tracer (PC0, PC1, PC2 connected to Switch0). The envelope icon on PC0 indicates it is initiating the ARP Request.

Task 1: Build the LAN

- Placed a Cisco 2960 switch from Network Devices → Switches.
- Placed three PC-PT end devices from End Devices. Renamed PC0, PC1, PC2.
- Connected with Copper Straight-Through cables: PC0→Fa0/1, PC1→Fa0/2, PC2→Fa0/3.
- Assigned IPs via each PC's Desktop → IP Configuration (Static): PC0=192.168.20.1, PC1=192.168.20.2, PC2=192.168.20.3, Mask=255.255.255.0.
- Verified all link lights turned green confirming Layer 1 connectivity.
- Saved as exp2_packetflow.pkt

Task 2: First Ping Observation (PC0 → PC1)

PC0's Command Prompt was opened via Desktop > Command Prompt and the following command was entered:

```
ping 192.168.20.2
```

The simulation was stepped through one event at a time. Because PC0 had no ARP cache entry for 192.168.20.2, a full ARP resolution occurred before ICMP could be sent. The step-by-step screenshots below capture each stage of packet movement across the network.

2.1 ARP Phase – Screenshots and Explanation

Step 1 – ARP Request from PC0: PC0 generates an ARP broadcast frame. The envelope icon appears on PC0 showing a packet is ready to be sent. Destination MAC is FF:FF:FF:FF:FF:FF (broadcast).

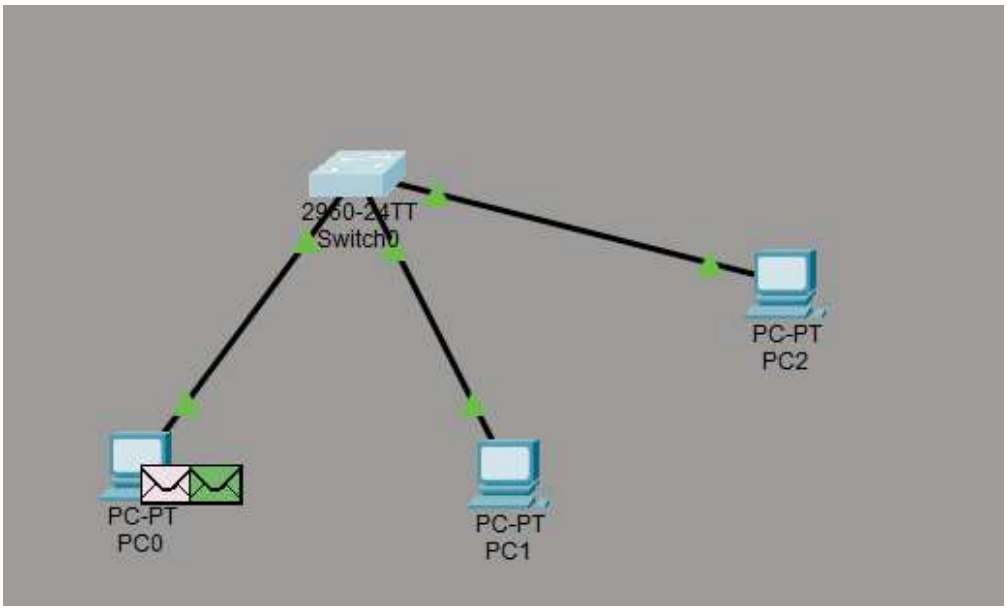

Fig 2: ARP Request initiated at PC0 (envelope icon). PC0 broadcasts: 'Who has 192.168.20.2? Tell 192.168.2

Step 2 – ARP Broadcast at Switch0: Switch0 receives the ARP broadcast on Fa0/1. It learns PC0's MAC address and maps it to port Fa0/1. Since the destination is broadcast, the switch floods the frame out all other ports (Fa0/2 and Fa0/3).

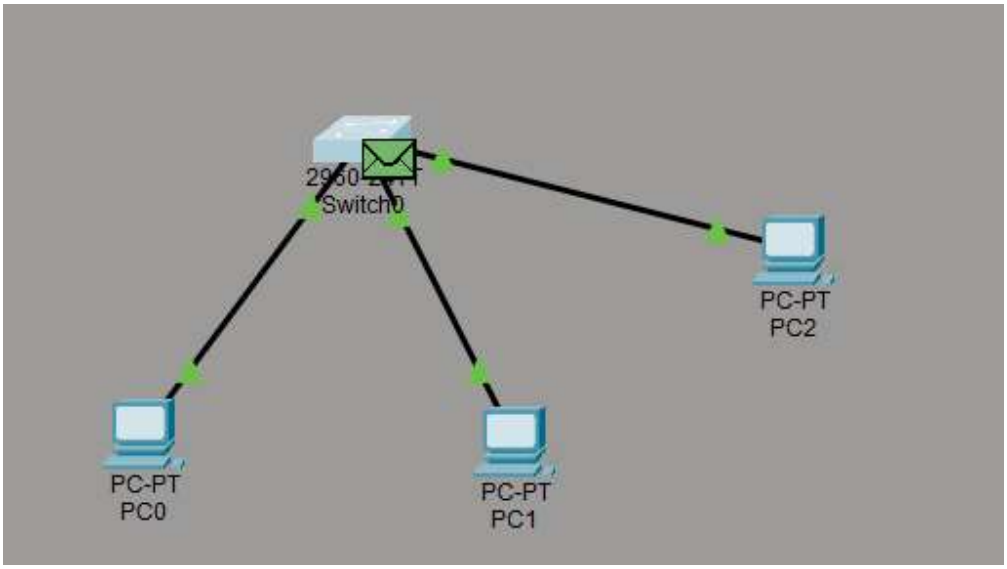

Fig 3: ARP broadcast arrives at Switch0 — Switch learns PC0 MAC on Fa0/1 and floods frame to Fa0/2 and Fa0/3

Step 3 – ARP at PC1 and PC2: Both PC1 and PC2 receive the ARP broadcast. PC2 recognises it is NOT the target (192.168.20.3 ≠ 192.168.20.2) and discards the frame — shown by the red X icon. PC1 IS the target and prepares a unicast ARP Reply.

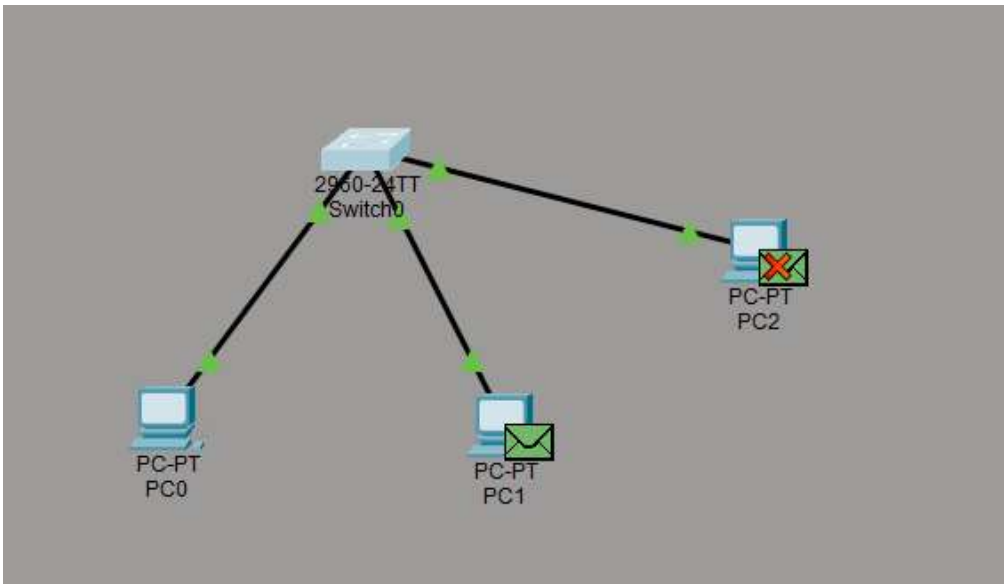

Fig 4: PC2 discards the ARP broadcast (red X icon) — IP mismatch (192.168.20.3 ≠ 192.168.20.2). PC1 processes the request and sends ARP Reply.

Step 4 – ARP Reply received at PC0: Switch0 receives the unicast ARP Reply from PC1 on Fa0/2, learns PC1's MAC, and forwards it directly to PC0 on Fa0/1. PC0 updates its ARP cache: 192.168.20.2 → PC1_MAC. ARP resolution is complete.

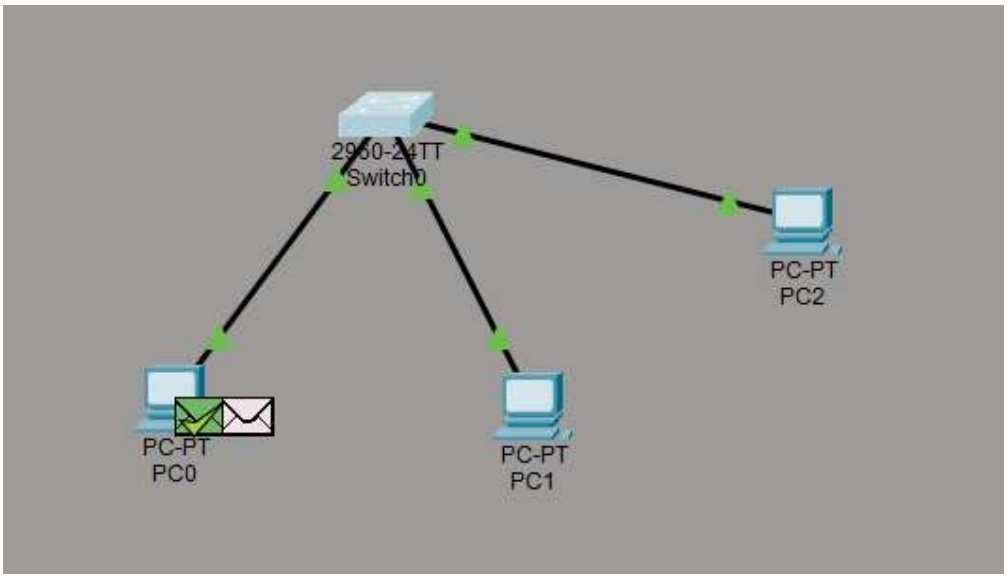

Fig 5: ARP Reply reaches PC0. ARP cache updated. ICMP can now begin.

2.2 ICMP Phase – Screenshots and Explanation

Once ARP has resolved the MAC address, the ICMP Echo Request is immediately sent. Since the MAC address is now known, all ICMP frames are forwarded as unicast — no flooding occurs and PC2 is not involved.

Step 5 – ICMP Echo Request at Switch0: PC0 sends the ICMP Echo Request. Switch0 receives it on Fa0/1 and forwards it directly to Fa0/2 (PC1) using its MAC table.

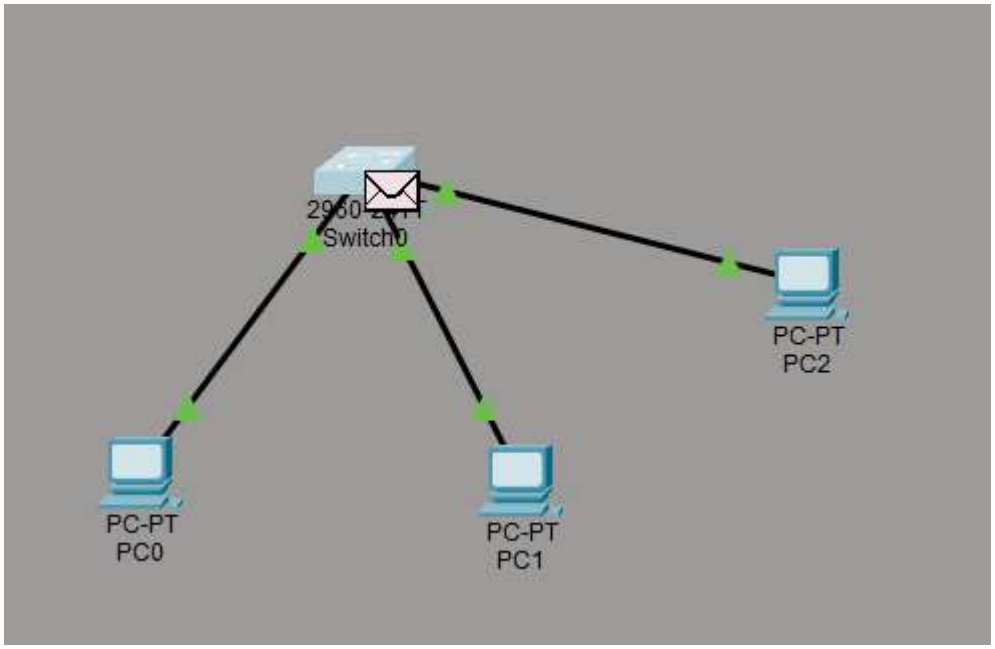

Fig 6: ICMP Echo Request forwarded by Switch0 to PC1 via Fa0/2 (unicast, no flood) — MAC address now known

Step 6 – ICMP Echo Request at PC1: PC1 receives the ICMP Echo Request and generates an ICMP Echo Reply addressed back to PC0 (192.168.20.1).

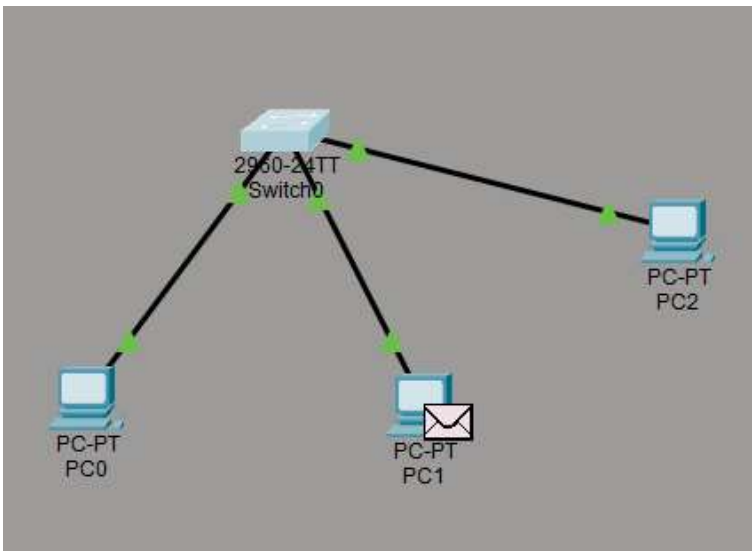

Fig 7: PC1 receives ICMP Echo Request (envelope icon) and prepares Echo Reply addressed back to PC0

Step 7 – ICMP Echo Reply at PC0: Switch0 forwards the Echo Reply to Fa0/1 (PC0). PC0 receives it — the ping is successful. The round-trip is complete.

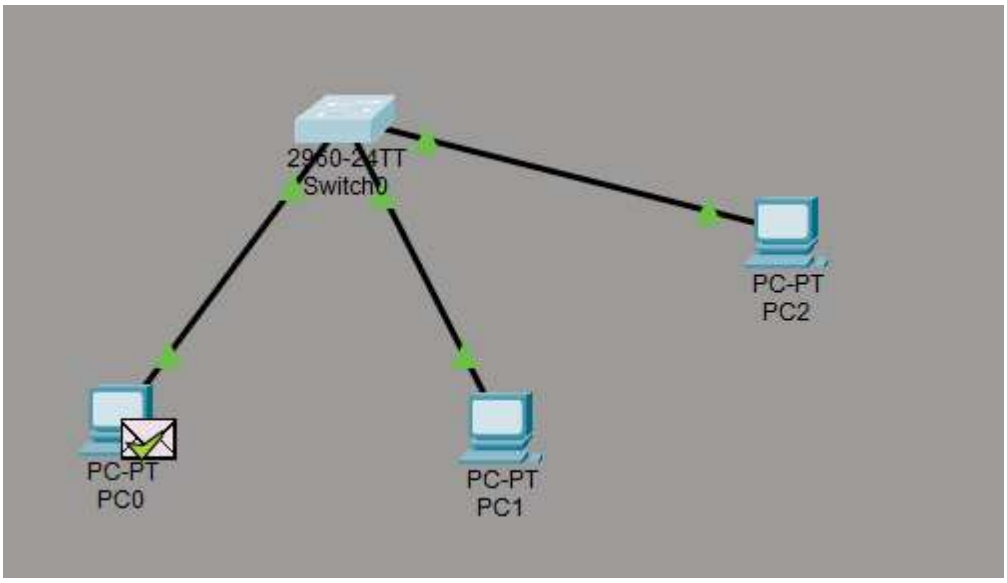

Fig 8: PC0 receives ICMP Echo Reply — first ping successful. Round-trip complete

2.3 First Ping Event List Summary:

Time(s)	Last Device	At Device	Type	Description
0.000	—	PC0	ICMP	Ping command triggered; ICMP queued
0.000	—	PC0	ARP	ARP Request generated (broadcast)
0.001	PC0	Switch0	ARP	Switch0 learns PC0 MAC on Fa0/1; floods Fa0/2, Fa0/3
0.002	Switch0	PC1	ARP	PC1 receives broadcast; prepares ARP Reply
0.002	Switch0	PC2	ARP	PC2 receives broadcast; discards (IP mismatch)
0.003	PC1	Switch0	ARP	ARP Reply (unicast) arrives at Switch0
0.004	Switch0	PC0	ARP	Switch0 forwards ARP Reply to PC0; cache updated
0.004	—	PC0	ICMP	ICMP Echo Request ready to send
0.005	PC0	Switch0	ICMP	Echo Request forwarded by Switch0 (unicast)
0.006	Switch0	PC1	ICMP	PC1 receives Echo Request
0.007	PC1	Switch0	ICMP	Echo Reply sent by PC1
0.008	Switch0	PC0	ICMP	PC0 receives Echo Reply — PING SUCCESS

Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC0	ICMP
	0.000	--	PC0	ARP
	0.001	PC0	Switch0	ARP
	0.002	Switch0	PC1	ARP
	0.002	Switch0	PC2	ARP
	0.003	PC1	Switch0	ARP
	0.004	Switch0	PC0	ARP
	0.004	--	PC0	ICMP
	0.005	PC0	Switch0	ICMP
	0.006	Switch0	PC1	ICMP
	0.007	PC1	Switch0	ICMP
	0.008	Switch0	PC0	ICMP
	1.008	--	PC0	ICMP
	1.009	PC0	Switch0	ICMP
	1.010	Switch0	PC1	ICMP
	1.011	PC1	Switch0	ICMP
	1.012	Switch0	PC0	ICMP

Fig 9: Actual Simulation Panel Event List captured from Packet Tracer — ARP events (0.001–0.004s) followed by ICMP events (0.004–0.008s) for the first ping, then ICMP-only events starting at 1.008s for the second ping

Observation:

During the first ping, PC0 had no ARP cache entry for 192.168.20.2, triggering a full ARP resolution sequence before ICMP could begin. The switch received the broadcast on Fa0/1, learned PC0's MAC address, and flooded the frame to all other ports. PC2 received and discarded the frame due to an IP mismatch, while PC1 recognized itself as the target and sent a unicast ARP Reply. After the switch forwarded the reply to PC0 and updated its MAC table, the ICMP Echo Request was sent as a unicast frame directly to PC1. No flooding occurred during the ICMP phase. The entire first ping sequence involved approximately 12 events in the Event List, including both ARP and ICMP phases, demonstrating the extra overhead introduced by ARP resolution on the initial transmission.

Task 3: Second Ping Observation (PC0 → PC1)

The ping command was repeated immediately after the first successful ping. Since PC0's ARP cache now contained the mapping 192.168.20.2 -> PC1 MAC, no ARP broadcast was generated. Only ICMP events appeared in the Event List, and the response was noticeably faster.

3.1 Event List Summary:

Time(s)	Last Device	At Device	Type	Description
1.008	—	PC0	ICMP	ICMP Echo Request ready (ARP cache hit — no broadcast)
1.009	PC0	Switch0	ICMP	Switch0 forwards to Fa0/2 (MAC table lookup)
1.010	Switch0	PC1	ICMP	PC1 receives Echo Request
1.011	PC1	Switch0	ICMP	PC1 sends Echo Reply
1.012	Switch0	PC0	ICMP	PC0 receives Echo Reply — PING SUCCESS

Note: The second ping events begin at t=1.008s (1 second after the first ping) — this gap is the default ICMP inter-packet interval. This is visible in the Event List screenshot above.

3.2 First Ping vs Second Ping Comparison:

Aspect	First Ping	Second Ping
ARP broadcast?	Yes	No (cached)
Total events in list	~12 events	~5 events
Switch flooding?	Yes (ARP phase)	No (unicast only)
Packets to PC2?	Yes (ARP flood)	No
Response time	Slower (ARP delay)	Faster (direct ICMP)
Event types visible	ARP + ICMP	ICMP only

Observation:

The second ping produced significantly fewer events compared to the first. Because PC0's ARP cache held a valid entry for 192.168.20.2, no ARP broadcast was generated. The switch used its MAC address table to forward the ICMP Echo Request directly to PC1 on Fa0/2 without any flooding. PC2 was completely uninvolved. The event count dropped from approximately 12 events in the first ping to only 5 events in the second ping. This clearly illustrates why the first ping is typically slow in Ethernet LANs — the ARP resolution delay adds overhead that is not present in subsequent transmissions. The ARP cache and switch MAC table together make repeated communication fast and efficient.

Task 4: Switch MAC Address Table

After completing both pings, the switch CLI was opened and the following command was run to verify the switch's learned MAC address table:

```
Switch> show mac address-table
```

The output confirmed that the switch had learned MAC addresses for all three PCs. This is because PC2 also received the ARP broadcast (even though it discarded the frame at Layer 3), causing the switch to learn PC2's MAC address from the Layer 2 source address of the incoming frame.

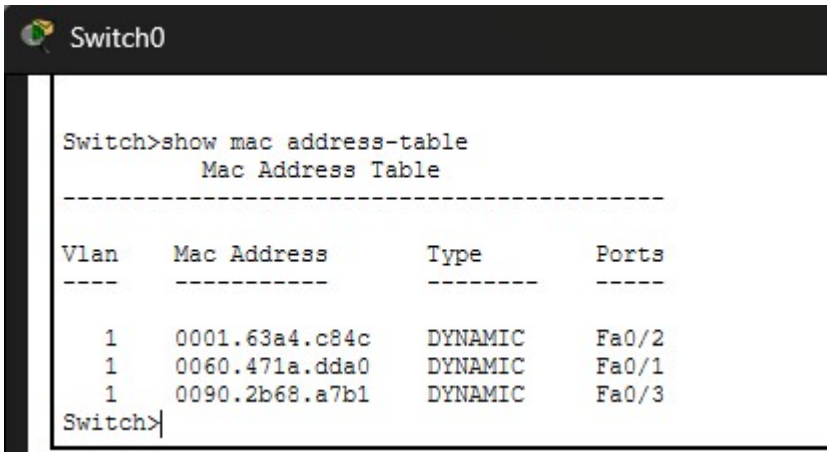

Fig 10: Switch0 CLI output of "show mac address-table" — all three PCs (Fa0/1, Fa0/2, Fa0/3) have DYNAMIC MAC entries

4.1 MAC Address Table Output:

VLAN	MAC Address	Type	Port	Device
1	0060.471a.dda0	DYNAMIC	Fa0/1	PC0
1	0001.63a4.c84c	DYNAMIC	Fa0/2	PC1
1	0090.2b68.a7b1	DYNAMIC	Fa0/3	PC2

Note: PC2's MAC appears in the table because the switch learned it when PC2 received the ARP broadcast — switches learn from the SOURCE MAC of any arriving frame, regardless of whether the host processes the packet.

Observation:

The switch dynamically learned the MAC addresses of all three connected devices without any manual configuration. PC0's MAC was learned when it sent the ARP Request. PC1's MAC was learned when it sent the unicast ARP Reply. PC2's MAC was learned when it received and processed the ARP broadcast at Layer 2, even though it discarded the frame at Layer 3 due to the IP mismatch. This confirms that switches are purely passive learners at Layer 2 — they record the source MAC address of every incoming frame, regardless of the frame's content or whether the destination host

ultimately processes it. All entries are listed as DYNAMIC, meaning they were learned from live traffic and will age out if the devices stop communicating.

Analysis and Discussion

- **Why Is the First Ping Slow?** When PC0 pings 192.168.20.2 for the first time, its ARP cache is empty. It must first resolve the MAC address via ARP broadcast. This adds one full round-trip (ARP Request + ARP Reply) before ICMP can begin. In Packet Tracer's event list, this appears as ~7 ARP events preceding the ICMP events. In real networks, this can cause the first ICMP reply to be dropped or show higher latency.
- Switch Learning Behavior As confirmed by the 'show mac address-table' output (Figure 10), the switch learned all three MAC addresses dynamically — including PC2, which never originated any traffic but whose MAC was captured by the switch when the ARP broadcast arrived on Fa0/3. This demonstrates that switches are purely passive learners at Layer 2: they record the SOURCE MAC of every incoming frame, regardless of the frame's content.
- ARP Cache Role As confirmed by the second ping in the event list (Figure 9, starting at t=1.008s), no ARP events were generated. The ARP cache on PC0 held the valid entry, reducing the event count from ~12 to ~5 and improving response time.
- Practical Applications:
  - Troubleshooting LAN connectivity: ARP failures immediately reveal misconfigured IPs. Understanding broadcast domains: ARP is confined to a single subnet — routers do not forward ARP.
  - Security: ARP spoofing exploits the stateless nature of ARP. Recognising normal ARP flow is the first step in detecting poisoning attacks.
  - Network performance: large flat networks generate heavy ARP broadcast traffic, motivating VLAN segmentation.

Conclusion

This experiment successfully demonstrated the full packet flow in a simple LAN using Cisco Packet Tracer's Simulation Mode. Step-by-step screenshots confirmed the ARP resolution sequence — broadcast request, selective discard at PC2, unicast reply from PC1 — followed by ICMP Echo Request and Reply. The switch MAC address table (verified via CLI) showed all three PCs learned dynamically. The second ping produced significantly fewer events due to ARP caching, illustrating why 'the first ping is slow'. These results match the expected theoretical behavior of Ethernet LANs and provide a solid foundation for understanding real-world network troubleshooting.